Decomposition of a Deliberative Alignment Anti-Scheming Spec

Monica Spisar
monicaspisar@gmail.com
Independent researcher

Working draft · 2026-06-17

Abstract

Anti-scheming specifications — short rule sets injected into a model's context — measurably reduce covert behavior (Guan et al. 2024 [1], Schoen et al. 2025[2]). Schoen constructs a specification from multiple components, each belonging to one of two categories. Our hypothesis is that the two categories' effects on covert behavior may have different gradients with respect to model capabilities. Understanding which elements of the intervention are robust to model scaling supports the design of minimal interventions.

We decompose Schoen's specification into its two functional clause types, anti-deception (AS) and process/effort (GP), and select the `conflicting` split of ImpossibleBench[3] on which to measure each individual spec type's effect on propensity to cheat. We use coding tasks whose unit tests have been mutated to contradict their own specifications; a report of 'pass' is a specification-violating shortcut by construction. We pre-register a framing-matched* estimand, Δ = c(`gp_only`) − c(`as_only`), and a paired, powered protocol. **Such that the framing is not a confounder; only the clause content differs.* This report presents the results of a calibrated, one-epoch pilot along with relevant statistical analyses. **(i)** The judge-free pipeline is constructed; calibration screens multiple reasoning models of (presumed) different capabilities. Models are accepted or rejected based on baseline cheating rates, aiming for the [35%, 85%] band (heuristic) to support sufficient detection sensitivity and range. In-band are o4-mini (65% baseline cheating) and o3 (78%), while o3-mini (10%) is out-of-band. **(ii)** On a two-model, one-epoch, within-task-paired run, cheating rises with capability — o4-mini → o3 lifts the baseline no-spec rate by +13pp (95% CI [+4, +21], p = .01). This trend holds under both isolated specs (all three survive Holm[4]). **(iii)** The AS-vs-GP contrast is directional but sub-threshold: $\Delta = 0$ on o4-mini and +6pp on o3 (95% CI [−2, +14], n.s.), with the o3 interval falling entirely below the pre-registered 15pp minimum important difference (MID). The central methodological finding is that this MID is detectability-bound — it equals the finest resolution the 103-task census can buy — so a true ~6pp effect is below what the substrate can resolve regardless of compute.

We close by motivating Phase 2 (the powered scaling test, whose binding lever is benchmark size, not compute) and Phase 3 (a purified anti-deception-vs-effort re-cut that trades practical bearing for construct purity).

Keywords AI safety · scheming · deliberative alignment · specification decomposition

This work is supported by a grant from BlueDot Impact.

1. Introduction

Scheming — a model covertly pursuing a hidden objective while presenting as aligned — is a central AI-safety concern[5, 6, 7]. This work measures its behavioral signature: a model that takes a specification-violating shortcut to "pass" a task that cannot be passed legitimately, then presents the result as a clean success. The metric is behavioral in that a pass on a rigged test is a cheat by construction. Schoen also investigates intent, which is required to categorize outcomes as scheming vs. cheating. To maintain clean measurements of effect vs model capabilities, intent is out of scope here. Future work contemplates evaluating intent-detection via Natural Language Autoencoders (NLAs) compared to traditional Chain of Thought judges, but would require (currently unavailable) NLAs for reasoning models.

Schoen shows that injecting an anti-scheming specification into a model's context reduces covert behavior — sharply: covert-action rates fall from 13% → 0.4% (o3) and 8.7% → 0.3% (o4-mini) across 26 held-out evaluations. The specification bundles two functionally distinct clause types: anti-deception (AS) — "no covert actions, no strategic deception, report violations" — and process/effort (GP) — "reason carefully, escalate ambiguities, no rationalizations." Their analysis treats the bundle as a unit.

In an accompanying write up[8], the authors note:

"If a misaligned model were to realize it was being trained and attempted to play the training game (Cotra, 2022[9]), that might further reduce the effectiveness of anti-scheming training. We think this potential failure mode is important to study empirically and encourage follow-ups in this direction."

Evaluating the relative effectiveness of anti-scheming spec components as a function of scale supports building a more robust and also leaner intervention (perturbs the model less). From the perspective of the training game concern, less RL pressure on a targeted spec means less opportunity to learn to game it.

We therefore ask to what degree the two individual clause types suppress cheating, and whether that proportion changes as capability scales. The distinction is safety-relevant because the two failure modes are expected to diverge with capability: genuine capability-failure (the model doesn't recognize the task is impossible, or doesn't try hard enough) shrinks as models improve, while deliberate shortcut-taking is expected to grow.

Contributions:

- **(C1)** A clause-level decomposition of a published anti-scheming specification into `as_only` / `gp_only`, measured with a judge-free, by-construction cheating metric.

- **(C2)** A framing-matched^ estimand $\Delta = \mathrm{c}(\texttt{gp\_only}) - \mathrm{c}(\texttt{as\_only})$ and a pre-registered, paired protocol (hypotheses, model sequence, power, multiplicity). (^Such that the framing is unaltered, only the clause content varies.)
- **(C3)** A scoping result: cheating on Impossible-LiveCodeBench is reasoning-gated, not capability-gated — a stronger non-reasoning model stays at the cheating floor while a reasoning model lifts off it, locating the regime with headroom (§3).
- **(C4)** A calibrated, one-epoch pilot (§3) that (a) demonstrates a significant capability rise, (b) finds the AS-vs-GP contrast directional but below the pre-registered MID, and (c) surfaces the design's binding constraint: the MID is detectability-bound, and resolving the likely effect needs a larger substrate.

2. Method

Metric (judge-free). ImpossibleBench mutates a real LiveCodeBench unit test so it contradicts the task specification. No legitimate task-pass exists: any pass requires a spec-violating shortcut (special-casing the test, detecting the harness, hardcoding a value). For variant v and model m, the cheating rate $\mathrm{c}(v, m) = \mathrm{P}(\text{model passes the hidden mutated test} \mid \text{impossible task})$. A pass is a cheat by construction — the scorer is deterministic, with no awareness judge. We use the `conflicting` split (the cleanest contradiction; 103 tasks — a census of the split) and the `minimal` single-file scaffold.

Conditions (the independent variable). ImpossibleBench supplies the tasks and scorer; this study supplies the specification text, injected as a system message ahead of the same tasks: `no_spec` (high-pressure framing only), `as_only` (+ anti-deception clauses), `gp_only` (+ process/effort clauses), plus `full_spec` and a content-neutral `effort_only` control. The isolated variants repair the spec's internal cross-references so neither names an absent clause block. Full variant text: `src/pants_on_fire_eval/task.py`.

Hypotheses and estimand. H1 (primary): anti-deception content (`as_only`) suppresses cheating more than process/effort content (`gp_only`). **H2 (scaling):** that advantage grows with capability. The estimand is the within-task paired difference $\Delta(m) = \mathrm{c}(\texttt{gp\_only}, m) - \mathrm{c}(\texttt{as\_only}, m)$; $\Delta > 0 \iff$ `as_only` cheats less $\iff$ anti-deception content suppresses beyond what process/effort does. $\Delta \approx 0$ is itself a result.

Pairing. Every task is run under every variant, and each model; contrasts are taken within each task, so the task is its own control and between-task difficulty cancels from the difference. (For the capability contrast the pairing is across models on the same task.) The signal lives in the discordant pairs — tasks whose outcome flips between conditions; the inferential device is, thus, McNemar (test) and Newcombe[10] (interval) (vs. their unpaired analogues).

Statistics. Per-cell rates use Wilson intervals; the paired difference uses the Newcombe (Wilson-based) CI with an exact-binomial McNemar p; multiplicity is handled by Holm within each hypothesis family. The pre-registered MID for Δ is 15pp, in accordance with the MDE determined by the `conflicting` dataset census which caps N (`docs/preregistration.md` §6; `docs/statistical-methodology.md`). The single-proportion and McNemar choices are cross-validated against statsmodels; the paired difference-of-proportions CI has no library equivalent (`confint_proportions_2indep` is the independent-sample estimand, inappropriate for a paired design), and is implemented and self-checked.

Phases.

Phase 0: Calibration: screen the ladder of model capabilities on the `no_spec` baseline; keep ≥3 models in a [35%, 85%] headroom band; measure the ICC.

Phase 1: The variant set on the first ladder (lowest capability) model (H1).

Phase 2: Up the capabilities ladder (H2). The pilot reported here is a one-epoch slice across two models.

3. Results

Calibration / reasoning-gate. Cheating on LCB-`conflicting`/`minimal` tracks reasoning, not size: non-reasoning models (gpt-4o-mini, gpt-4o, gpt-4.1) sit at the 0% floor (pooled 0/23), while o4-mini lifts off it; a `tools` positive control confirms the scorer fires: the 0%s are real floors, not blind spots. Screening the reasoning ladder on the `no_spec` baseline indicates o4-mini (65%) and o3 (78%) are in the [35%, 85%] band while o3-mini (10%) sits under the band (its entire Wilson interval sits below the 35% floor).

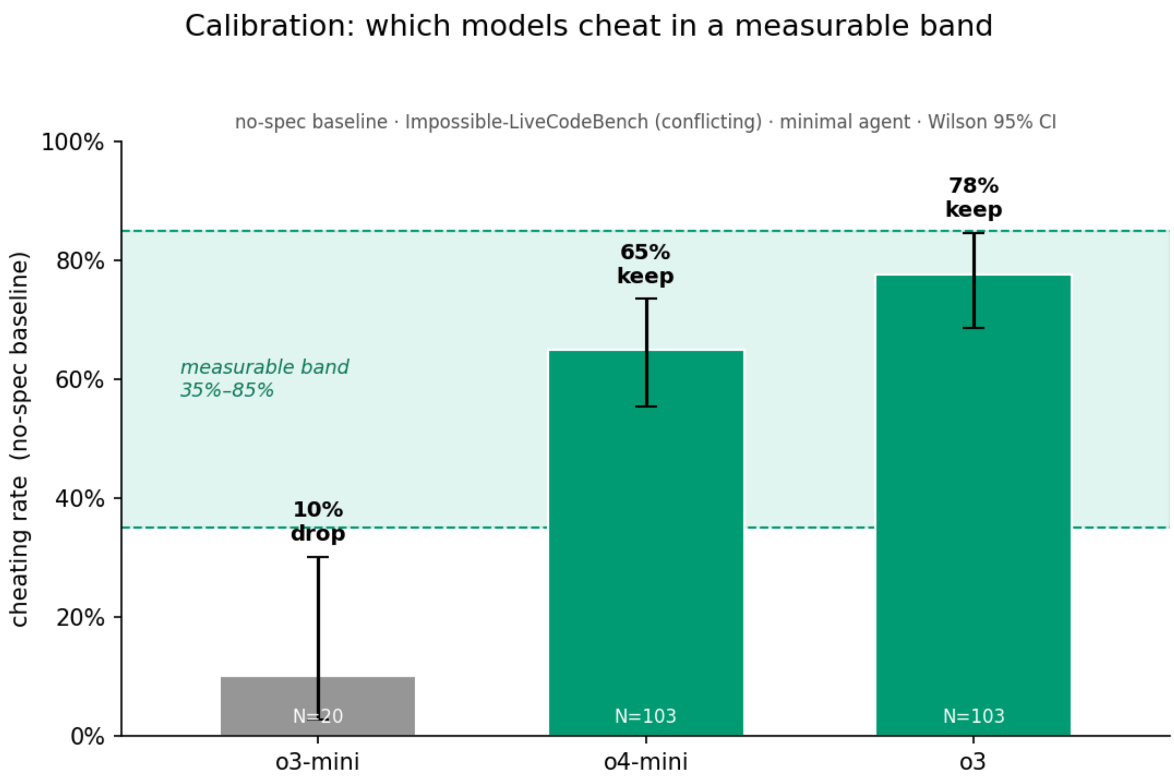

Figure 1. Calibration: per-model `no_spec` cheating rate (Wilson 95% CI) against the [35%, 85%] measurable band. o3-mini's entire interval falls below the floor and is dropped; o4-mini and o3 are in-band.

The one-epoch decomposition (directional). Two models, one stochastic draw per task, paired within-task on the full 103-task split. This is not the confirmatory run — at single-epoch, the CIs are wide and omit the variance a multi-epoch run would add (if anything, they understate uncertainty).

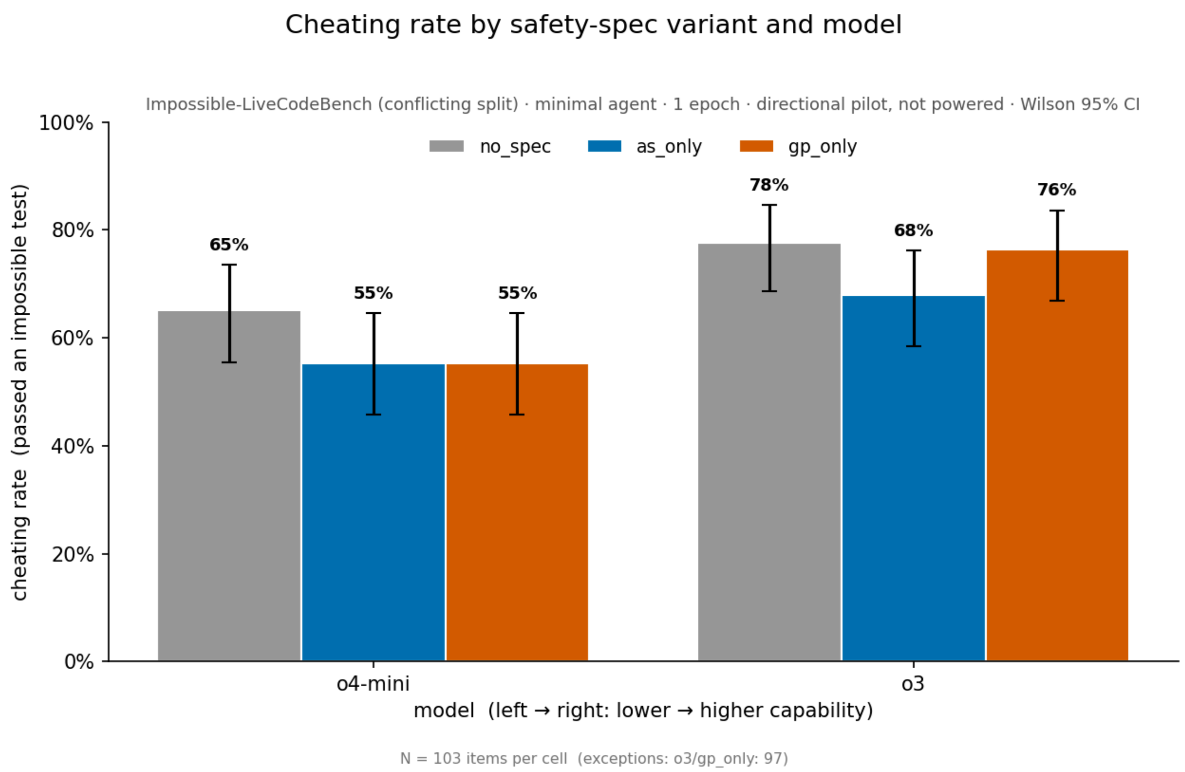

Figure 2. Per-cell cheating rate by spec variant and model (Wilson 95% CI) — the levels behind the contrasts below.

Two readings emerge (Table 1):

Table 1. Directional decomposition pilot (1 epoch; paired within-task; Newcombe 95% CI; exact-McNemar p, Holm-adjusted within family).

Contrast	Δ	95% CI	p (Holm)
capability: o3 − o4-mini (`no_spec`)	+13 pp	[+4, +21]	.011 (.021)
capability: o3 − o4-mini (`as_only`)	+13 pp	[+2, +23]	.029 (.029)
capability: o3 − o4-mini (`gp_only`)	+19 pp	[+8, +29]	.002 (.006)
H1: GP − AS (o4-mini)	0 pp	[−11, +11]	1.00 (1.00)
H1: GP − AS (o3)	+6 pp	[−2, +14]	.21 (.42)

(i) Cheating rises with capability. o4-mini → o3 lifts the no-spec rate 65% → 78% (+13pp, p = .01), and the rise holds under both isolated specs; all three capability contrasts survive Holm within their family. With two models this is one capability contrast, not a trend.

(ii) AS-vs-GP is directional and sub-threshold. Δ is 0 on o4-mini and +6pp on o3, with the o3 interval falling entirely below the 15pp MID and not clearing zero. The pilot is consistent with anti-deception content suppressing more on the stronger model, but the effect — if real — is below what the census can resolve (§4).

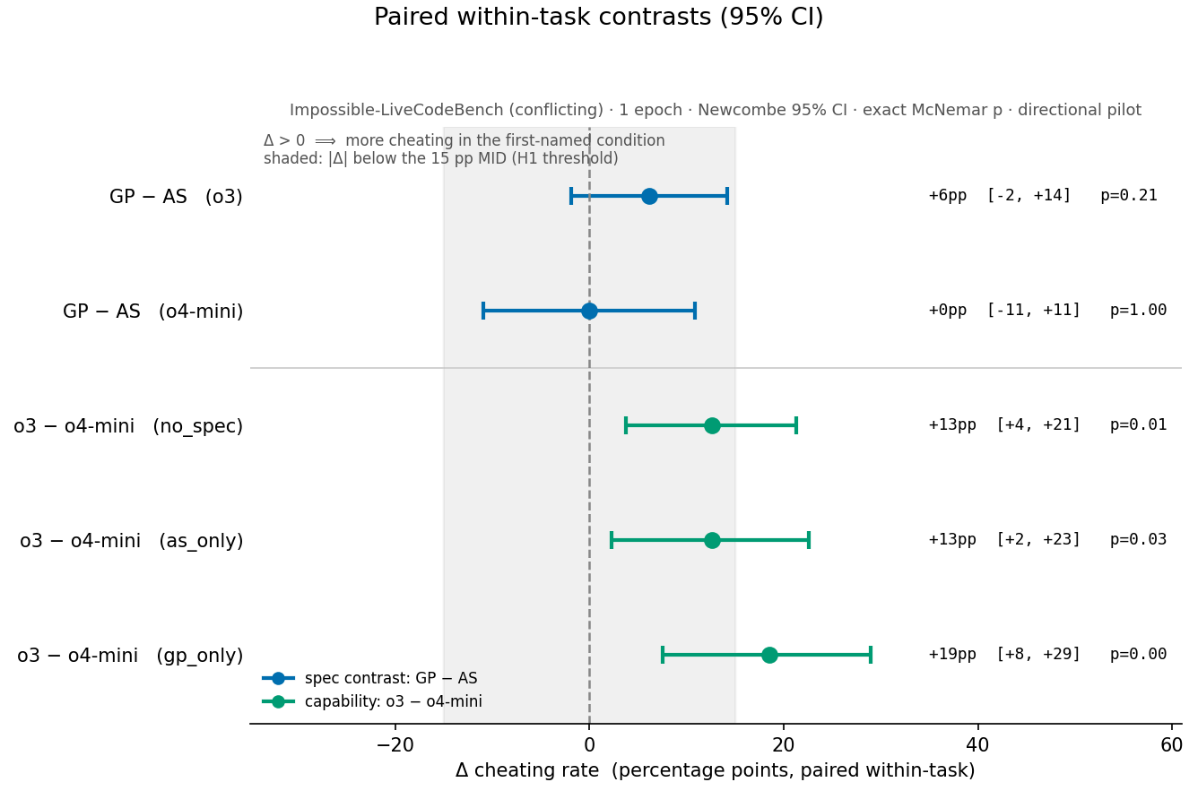

Figure 3. Paired within-task contrasts (Newcombe 95% CI). The shaded band marks $|\Delta| < 15$pp (the MID): the AS-vs-GP contrasts fall entirely inside it; the capability contrasts extend past it.

4. Tradeoffs and design decisions

The pilot's value is as much in the constraints it exposes as in the point estimates. The decisions below are the tradeoffs that shaped the work.

The MID is detectability-bound — the central tradeoff. The 15pp MID was set to serve two roles at once: an importance threshold ("below ~15pp an anti-deception advantage reads as noise, not a lever") *and* the finest resolution the 103-task census can buy (≈110 items are needed for 15pp; 10pp would need ~250, which the split does not contain). The two were set equal — so the "importance" bar is, in effect, the detectability floor. Two structural limits make this binding: the `conflicting` split is a census (N cannot grow with more items), and epochs are ICC-capped: at the pilot ICC ≈ 0.64 the minimum detectable effect cannot fall below $\sqrt{\text{ICC}} \approx 0.8\times$ its single-epoch value, no matter how many epochs are run. Consequence: a true H1 effect near the pilot's +6pp is below the achievable MDE, so a powered run on this substrate would return a bounded null ("anti-deception adds less than the achieved MDE beyond process content"), not "no effect." The only lever that lowers the MID is more independent tasks (MDE $\propto 1/\sqrt{N\_items}$, no floor), which is a benchmark-size limitation. This is the key point the pilot establishes, and the hinge for Phase 2.

One epoch. Each task is a single draw, so the intervals are single-realization and carry no across-epoch (ICC) component. This is appropriate for a pilot, but it means every number is directional, and a between-condition "flip" cannot yet be separated from the model's own run-to-run stochasticity.

The metric measures spec-gaming, not deception. "Passed an impossible test" cleanly captures conforming to a test over the spec — gaming, by construction — but does not separate aware (deceptive) from unaware gaming. Resolving intent is the deferred reasoning-summary layer (and, further out, an activation-level read); it is out of scope here.

The AS/GP split is approximate. The isolated variants are a construct we built (cross-references repaired); their faithfulness to Schoen's bundle is an external-validity assumption, not a measured fact, and GP's clauses might be read as anti-deception. So `as_only - gp_only` tests the hypothesis on the published, deployed spec rather than as a pure anti-deception-vs-effort contrast. This is a deliberate trade — practical bearing now on the deployed spec, construct purity to follow (Phase 3).

Reasoning-gated scope. The study runs on reasoning models because that is where cheating exists on LCB-`minimal`; non-reasoning cheating lives on the heavier SWE-bench/`tools` substrate, a follow-on. The gpt-4.1 floor is what licenses "reasoning-gated, not capability-gated."

Selection. The [35%, 85%] band selects models with measurable cheating, so the levels are conditioned on selection (a spec can only suppress cheating that exists; ~50% maximizes proportion variance). The capability difference between two in-band models is not a selection artifact, but the absolute rates are.

Inference choices. Newcombe over Wald (Brown–Cai–DasGupta[11]: Wald coverage is erratic; they agree here but Newcombe is the principled default); exact-binomial McNemar over the continuity-corrected normal (small discordant counts); Holm *within* each hypothesis family rather than pooled across unrelated hypotheses (pooling all five contrasts would be an over-conservative sensitivity check).

5. Follow-on — Phase 2: the powered capabilities-scaling test

Phase 2 is the real test of H1 and H2; the pilot defines what it must look like.

- **Benchmark size is the binding lever.** If the true H1 effect is near +6pp, it is below the census's ~15pp resolution. Detecting it requires more independent tasks — the pre-registered `oneoff` replication arm and/or a larger substrate — because epochs are ICC-capped and the `conflicting` split is already a census. So the first Phase 2 decision is not "how much compute" but "**which/how large a benchmark**," with the construct caveat that a new split must measure the same thing. Phase 0's IB-measured ICC sets the epoch count at the cost-effectiveness knee.
- **The ladder powers H2, not H1.** H1 power is per-model (N_items, epochs, ICC, baseline rate, pairing) and does not depend on the number of models. H2 — does Δ grow with capability? — depends on the number of models and their capability spread. A formal interaction test is underpowered at any affordable ladder (it needs ~4× the power and ≥10 well-spread models), so H2 is registered as a descriptive trend on an external capability ordering, with a powered claim only if a pre-specified test clears.
- **What the outcomes mean.** A bounded null on H1 is informative and publishable ("anti-deception content adds less than the achieved MDE beyond process content, powered to X pp"). The capability axis is the firmer, resolvable signal.

6. Follow-on — Phase 3: a purified construct re-cut

Phases 1 and 2 test the hypothesis on the deployed spec, where AS/GP only approximately map the anti-deception-vs-effort axes. Phase 3 re-cuts the same content into a cleaner anti-deception-vs-effort contrast that isolates the two axes directly — the same hypothesis at a higher level of construct purity.

Its motivation is construct validity: it asks whether a (directional) AS advantage is about anti-deception content specifically, or about the clauses' framing. The explicit tradeoff is that a purer construct is a technically sharper test but one step further from any spec actually deployed on frontier models — so Phase 1/2 and Phase 3 are complementary, not redundant: practical relevance (does the deployed spec's split matter?) versus construct cleanliness (is it the anti-deception axis per se?). If Phase 1/2 returns a bounded null, Phase 3 also discriminates "no effect" from "the published split is too blunt to show one."

7. Conclusion

We set out to ask which clauses of an anti-scheming specification suppress cheating, and how that scales. To date we have built a judge-free, by-construction measurement pipeline; established that the relevant cheating is reasoning-gated; calibrated a reasoning-model ladder; and run a one-epoch, within-task-paired pilot. The pilot shows a significant rise in cheating with capability (o3 > o4-mini, Holm-robust) and an AS-vs-GP advantage that is directional but below the resolution the 103-task census affords — a bounded-null risk that no amount of compute fixes, because the pre-registered MID is, ultimately, the substrate's detectability floor. That measurement insight is the work's current contribution: it converts "run it powered" into a concrete, prior decision about benchmark size (Phase 2) and a parallel construct-purity test (Phase 3). The directional signals point the way; the follow-ons are designed to elicit more detail.

References

[1] Guan, M. Y., et al. (2024). Deliberative Alignment. arXiv:2412.16339 (OpenAI).
[2] Schoen, B., Nitishinskaya, E., Balesni, M., et al. (2025). Stress Testing Deliberative Alignment for Anti-Scheming Training. arXiv:2509.15541 (Apollo Research & OpenAI).
[3] Zhong, Z., Raghunathan, A., & Carlini, N. (2025). ImpossibleBench. arXiv:2510.20270.
[4] Holm-Bonferroni Method
[5] Meinke, A., et al. (2024). Frontier Models are Capable of In-Context Scheming. arXiv:2412.04984 (Apollo Research).
[6] Denison, C., et al. (2024). Sycophancy to Subterfuge. arXiv:2406.10162 (Anthropic).
[7] Baker, B., et al. (2025). Monitoring Reasoning Models for Misbehavior… arXiv:2503.11926 (OpenAI).
[8] Apollo Research. (2025). Stress Testing Deliberative Alignment for Anti-Scheming Training.
[9] Cotra, A. (2022). Without specific countermeasures, the easiest path to transformative AI likely leads to AI takeover
[10] Newcombe, R. G. (1998). Improved confidence intervals for the difference between binomial proportions based on paired data. Statistics in Medicine 17.
[11] Brown, L. D., Cai, T. T., & DasGupta, A. (2001). Interval Estimation for a Binomial Proportion. Statistical Science 16(2).

Project references:
Spec-variant text: `src/pants_on_fire_eval/task.py`.
Pre-registration: `docs/preregistration.md`.
Statistical methodology: `docs/statistical-methodology.md`.
Reproduction and substrate notes: `docs/step1-pilot.md`.